

Cyril Creque-Sarbinowski

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■ External Resource

■ Slides/Notes

■ Paper

Education

Johns Hopkins University (JHU)

Ph.D in Physics & Astronomy

Advisor: Marc Kamionkowski

Baltimore, MD

2017-2022

Massachusetts Institute of Technology (MIT)

B.S. in Physics & Mathematics

Cambridge, MA

2013-2017

Positions Held

Center for Computational Astrophysics (CCA)

Flatiron Research Fellow

New York, NY

August 2022-Present

New York University (NYU)

Visiting Scholar

New York, NY

September 2022- September 2024

Awards, Scholarships, Fellowships

National Science Foundation

Graduate Research Fellow

2019-2022

Gates Millenium Scholars Program

Scholar

2013-2022

JHU Kelly Miller Fellowship

Fellow

2017-2019

Teaching/Outreach

Sound Plus Science

Chief Facilitator & Mentor, Developed Curriculum [Sample Exercise]

Civic Hall

March 2024-Present

GEM Annual Conference

Flatiron Institute Booth Staff

September 2023, 2024

ASTRO 10000: Basic Concepts in Astronomy

Guest Lecturer

Hunter College

May 2023

PHYS 111: Introductory Physics

Virtual Guest Lecturer

Moravian University

Spring 2022

PHYS2300: Quantum Theory of Fields I

Guest Lecturer

Brown University

Spring 2022

AS.171.107 General Physics for Physical Science Majors

Teaching Assistant

JHU

Fall 2017

Lecturer: Prof. Robert Leheny & Prof. Rosemary Wyse

AS.173.111 General Physics Laboratory I

Section Head

JHU

Fall 2017

Selected Conferences, Workshops, Seminars

* = *Invited*, † = *Organizer*

2024.....		
Workshop	Cosmology in the Adriatic – From PT to AI , BCCP, Ruder Bošković Institute, Faculty of Science in Split	<i>July</i>
<i>Seminar*</i>	Cosmology Group Meeting, Perimeter Institute	<i>July</i>
Program	Dark Matter Theory, Simulation, & Analysis in the Era of Large Surveys , Kavli Institute for Theoretical Physics (KITP), Affiliate	<i>May-June</i>
Workshop	The Future of Dark Matter , Columbia University/ Flatiron Institute	<i>April</i>
Symposium†	Caribbean Future of Science Symposium	<i>March</i>
<i>Seminar*</i>	KICP Seminar , University of Chicago	<i>January</i>
2023.....		
<i>Workshop*</i>	Large-Scale Parity Violation Workshop , Academia Sinica Institute of Astronomy & Astrophysics (ASIAA)	<i>December</i>
Workshop†	Parity Violation from Home	<i>October</i>
<i>Talk*</i>	Cosmology Talks	<i>August</i>
<i>Workshop*</i>	Ending Inflation & the Hot Big Bang , Stonybrook University	<i>June</i>
<i>Workshop*</i>	Modified Gravity Program , Illinois Center for Advanced Studies of the Universe	<i>May</i>
<i>Symposium*</i>	Future Faculty Symposium in Physics , Princeton University	<i>April</i>
<i>Seminar*</i>	Quantum Gravity Seminar , Perimeter Institute	<i>April</i>
<i>Seminar*</i>	Cosmology Seminar , Max Planck Institute for Astrophysics	<i>April</i>
<i>Seminar*</i>	Cosmology Seminar , Tufts/MIT	<i>February</i>
<i>Seminar*</i>	Brown University IDEA Seminar	<i>February</i>
2022.....		
<i>Conference</i>	NSBP Annual Conference	<i>November</i>
<i>Workshop*</i>	Guest Expert, Cosmology Talks Parity-Violation Mini-Workshop	<i>November</i>
<i>Seminar*</i>	Theory Group Seminar , Imperial College London	<i>November</i>
<i>Talk*</i>	AstroParticle Lunch , The Ohio State University	<i>September</i>
<i>Talk</i>	Early Universe/Cosmology Lunch Discussion , Princeton University/Institute of Advanced Study	<i>September</i>
<i>Workshop†</i>	NeXus Workshop , CCA, Flatiron Institute	<i>August</i>
Workshop	Large-Scale Structure Cosmology beyond 2-Point Statistics , Aspen Center for Physics	<i>May-June</i>
<i>Seminar*</i>	Brown University	<i>March</i>
$t \leq 2021$		
Workshop	NeXus Workshop , CCA, Flatiron Institute	<i>June 2021</i>
Conference	Phenomenology 2021	<i>May 2021</i>
Workshop	XIX International Workshop on Neutrino Telescopes	<i>February 2021</i>
Conference	Neutrino 2020	<i>June 2020</i>
<i>Seminar*</i>	Physics Seminar , Bowdoin College	<i>November 2019</i>
Conference	NSBP Annual Conference	<i>November 2019</i>
Conference	Cosmic Controversies , Kavli Institute for Cosmological Physics (KICP)	<i>October 2019</i>
Symposium	Symposium in Honor of the Legacy of Vera Rubin , Georgetown University	<i>June 2019</i>
Workshop	Novel Ideas for Dark Matter , Princeton Center for Theoretical Science (PCTS)	<i>January 2019</i>
<i>Conference</i>	NSBP Annual Conference	<i>November 2018</i>

Conference	Identification of Dark Matter, Brown University	July 2018
Workshop	Second Annual Line-Intensity Mapping Workshop (IM@Hopkins), Johns Hopkins University (JHU)	July 2017
Conference	Neutrinos & Light Particles in Cosmology, Berkeley Center for Cosmological Physics (BCCP)	June 2016
Conference	NSBP Annual Conference	February 2015

Professional Activities

Referee	Phys. Rev. D (2023–), Phys. Rev. Lett. (2023–) MNRAS (2022–), Philos. Trans. R. Soc. A (2024–)	
Leader	NeXus	2022–Present
Organizer	Particles Vs. New Probes (P Vs. NP), Lead Organizer	March 2025
	Caribbean Future of Science Symposium, SOC	March 2024
	Parity Violation from Home, Co-Organizer	October 2023
	NeXus Workshop, Lead Organizer	August 2022
Review	GEM Fellowship Program	November 2023
Panelist		

Publications [13 ref. journal pub. (10 1st-author) +1 submitted, 300+ citations, h-index: 8]

- [1] S. Alexander, H. Bernardo, and **C. Creque-Sarbinowski**, “Nontriviality of dynamical Chern-Simons gravity and the standard model,” *Phys. Rev. D* **110** no. 2, (2024) 025015, [arXiv:2403.15657 \[hep-th\]](#).
- [2] S. Alexander, **C. Creque-Sarbinowski**, H. Gilmer, and K. Freese, “Higgs Inflation and the Electroweak Gauge Sector,” [arXiv:2306.04671 \[hep-ph\]](#).
- [3] J. Flitter, **C. Creque-Sarbinowski**, M. Kamionkowski, and L. Dai, “Magnetic fields from compensated isocurvature perturbations,” *Phys. Rev. D* **107** no. 10, (2023) 103536, [arXiv:2304.03299 \[astro-ph.CO\]](#).
- [4] **C. Creque-Sarbinowski**, S. Alexander, M. Kamionkowski, and O. Philcox, “Parity-violating trispectrum from Chern-Simons gravity,” *JCAP (20th Anniversary Special Issue)* **11** (2023) 029, [arXiv:2303.04815 \[astro-ph.CO\]](#).
- [5] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Subtracting compact binary foregrounds to search for subdominant gravitational-wave backgrounds in next-generation ground-based observatories,” *Phys. Rev. D* **108** no. 6, (2023) 064040, [arXiv:2209.01310 \[gr-qc\]](#).
- [6] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Compact Binary Foreground Subtraction in Next-Generation Ground-Based Observatories,” [arXiv:2209.01221 \[gr-qc\]](#).
- [7] S. Alexander and **C. Creque-Sarbinowski**, “UV completions of Chern-Simons gravity,” *Phys. Rev. D* **108** no. 10, (2023) 104046, [arXiv:2207.05094 \[hep-ph\]](#).
- [8] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “High-energy astrophysical neutrinos from cosmic strings,” *Phys. Rev. D* **107** no. 12, (2023) 123019, [arXiv:2206.06377 \[hep-ph\]](#).
- [9] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “Seeking neutrino emission from AGN through temporal and spatial cross-correlation,” *Phys. Rev. D* **105** no. 12, (2022) 123035, [arXiv:2111.08012 \[astro-ph.HE\]](#).

- [10] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “AGN variability in the age of VRO,” *Astrophys. J.* **941** (2022) 41, [arXiv:2110.13149 \[astro-ph.GA\]](#).
- [11] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “Resonant neutrino self-interactions,” *Phys. Rev. D* **103** no. 2, (2021) 023527, [arXiv:2005.05332 \[hep-ph\]](#).
- [12] **C. Creque-Sarbinowski**, L. Ji, E. D. Kovetz, and M. Kamionkowski, “Direct millicharged dark matter cannot explain the EDGES signal,” *Phys. Rev. D* **100** no. 2, (2019) 023528, [arXiv:1903.09154 \[astro-ph.CO\]](#).
- [13] **C. Creque-Sarbinowski** and M. Kamionkowski, “Searching for Decaying and Annihilating Dark Matter with Line Intensity Mapping,” *Phys. Rev. D* **98** no. 6, (2018) 063524, [arXiv:1806.11119 \[astro-ph.CO\]](#).
- [14] **C. Creque-Sarbinowski**, S. Bird, and M. Kamionkowski, “Cross-correlation between thermal Sunyaev-Zeldovich effect and the integrated Sachs-Wolfe effect,” *Phys. Rev. D* **94** no. 6, (2016) 063519, [arXiv:1606.00839 \[astro-ph.CO\]](#).

References

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